MUHAMMAD SHAYAN AYUB

Full Stack Developer | Backend Engineering | Cybersecurity

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PROFESSIONAL SUMMARY

Computer Science graduate and Full Stack Web Developer with professional experience developing web applications, integrating APIs, implementing authentication functionality, and building backend services. Experienced with Python, FastAPI, React.js, PostgreSQL, Docker, Redis, Celery, WebSockets, and Tailwind CSS. Practical cybersecurity background includes PCAP-based malware traffic investigation, HTTP security header analysis, OWASP Top 10 concepts, and hands-on laboratory exercises. Seeking international opportunities in full-stack engineering, backend development, or software engineering.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), C++, SQL, HTML5, CSS3, PHP

Backend & API Engineering: FastAPI, RESTful APIs, JWT Authentication & Authorization, WebSockets, Celery, Microservices Patterns, AsyncIO

Frontend Development: React.js, Tailwind CSS, Responsive Web Design, Component Architecture, State Management, API Integration

Databases & Data Management: PostgreSQL, MySQL, SQLite, Relational Modeling, Schema Optimization, pgAdmin

Caching & Background Tasks: Redis, Celery Task Queues

DevOps & Infrastructure: Docker, Git, GitHub, Linux (Debian/Kali), VS Code, Postman

Cybersecurity & Network Analysis: OWASP Top 10 Mitigation, HTTP Security Headers, Wireshark, PCAP Forensics, NetworkMiner, Nmap Fundamentals

Cloud Fundamentals: Microsoft Azure Concepts (Governance, Cost Management, Monitoring, Security Fundamentals)

PROFESSIONAL EXPERIENCE

Full Stack Web Developer — HexaForge

June 2025 – Present | Pakistan

Collaborative, project-based engineering environment delivering full-stack products and backend APIs

- Engineered full-stack web applications in an agile environment, bridging client interfaces with scalable backend infrastructure.
- Developed and maintained performant RESTful APIs and WebSocket endpoints using Python and FastAPI for asynchronous operations.
- Implemented secure token-based authentication (JWT) and authorization mechanisms across web application workflows.
- Structured, queried, and administered relational databases with PostgreSQL, optimizing schema designs and data integrity.
- Built responsive, component-driven user interfaces utilizing React.js and Tailwind CSS with cross-device compatibility.
- Containerized microservices and development workflows using Docker to ensure environment parity and reproducible deployments.
- Integrated Redis caching and Celery task queues to handle asynchronous background processing and improve system throughput.

SELECTED SOFTWARE DEVELOPMENT PROJECTS

Digital Assets Marketplace *React.js, FastAPI, PostgreSQL, Docker*

- Engineered a decoupled full-stack digital asset platform designed for listing, catalog discovery, and transactional asset workflows.
- Designed modular backend services with FastAPI, enforcing schema validation, role access, and relational constraints in PostgreSQL.
- Containerized frontend and backend services using Docker, facilitating multi-container local orchestration and deployment consistency.

Student Issue Management System *Python, FastAPI, JavaScript, PostgreSQL | GitHub*

- Built an institutional ticket-tracking and issue resolution platform for university administrative and student workflows.
- Developed RESTful lifecycle endpoints in FastAPI to support issue logging, escalation tracking, and administrative resolution states.
- Integrated PostgreSQL relational schemas ensuring tamper-evident history, relational integrity, and reliable status persistence.

Islamabad NGO Donation Website *PHP, MySQL, HTML5, CSS3, JavaScript | GitHub*

- Created a donation management web platform supporting transparent philanthropic tracking and orphan support campaigns.
- Implemented relational donor records, administrative contribution logs, and responsive contribution dashboards in MySQL and PHP.

Mini Blog Application *JavaScript, HTML5, CSS3 | GitHub*

- Engineered a responsive web application for creating, managing, and categorizing publishing content with client-side state handling.

CYBERSECURITY EXPERIENCE & PRACTICAL TRAINING

CyberDefenders — DanaBot Digital Forensics Lab *April 2025 | Verification Record*

- Executed a comprehensive PCAP network traffic forensic investigation to isolate malware behavior and suspicious host interactions.
- Analyzed packet captures using Wireshark and NetworkMiner to reconstruct command-and-control (C2) communication patterns.
- Correlated extracted binary payloads and network hashes with threat intelligence sources (VirusTotal, ANY.RUN).

TryHackMe — Hands-On Cybersecurity Learning *2025 | Platform Profile*

- Completed 22+ interactive rooms covering ethical hacking concepts, network enumeration, privilege escalation, and web security.
- Achieved a reported platform ranking placing within the top 25% of active global learners.

National Cybersecurity Training Program — MUST *2025 | Mirpur University of Science & Technology*

- Completed practical training covering SOC fundamentals, defensive countermeasures, network traffic analysis, and security controls.

CERTIFICATIONS & CONTINUOUS LEARNING

- **HackerRank Skill Certifications (2025):** Verified skill certifications in C++ Problem Solving and MySQL Database Operations (Verification).
- **Microsoft Learn Cloud Achievements (2024–2025):** Completed foundational learning modules covering Azure Architecture, Governance, Compliance, Monitoring, Cost Management, and Cloud Security (Verification).

EDUCATION

Bachelor of Science in Computer Science (BSCS) *Graduation: October 2026*

Mirpur University of Science and Technology (MUST), Pakistan | Official DMC / transcript pending formal issuance

Core Disciplines: Data Structures & Algorithms, Database Systems, Computer Networks, Information Security, Operating Systems & Linux Architecture, Software Engineering Methodologies.